

Design Space Exploration of Verifiable Credential Schemas using Partial Graph Modeling

Martin Farkas, Bertalan Zoltán Péter, Imre Kocsis

Abstract: With the advent of the use of standardized verifiable credentials, information verifiers begin to face a design space exploration problem: what kinds and potential sets of verifiable claims to request from persons and organizations to meet their verification needs, especially so that on the one hand, they have to maximize the trustworthiness of claim verification, but on the other hand, also observe the principle of data minimization and the correlation risk minimization needs of credential holders. We propose approaching this design problem through partial graph modeling and generation. We demonstrate that partial graph modeling can be amenable to express the potentially diverse information graph requirements of a verification scenario and generate admissible templates for so-called verifiable presentation graphs. This facilitates the application of established design space exploration approaches, including the enumeration of options integrated with multi-aspect evaluation and rank ordering.

Keywords: verifiable credentials, information graph, partial graph modeling

1 Introduction

In recent years, verifiable credentials (VCs) [1, 2] have stirred the interest of not just researchers but also legislators and corporations, as the European Union decided to adopt the ESSIF framework [3]. They are a crucial part of a novel and proliferating identity model called self-sovereign identity, which aims to offer its users more interoperability, privacy, and control over their identity on the Internet. VCs represent claims, and as more and more interdependent ecosystems and policies utilize them and the complexity of the claims grows, their design rapidly becomes non-trivial.

The following question arises: How do we maximize the expressive power of a related set of verifiable credentials while maintaining their holder's control over their claims by minimizing the amount of data shared and the risk of correlation?

To provide an answer to this question, we propose a partial-graph-modeling-based design space exploration workflow using the principles of model-driven architectures.

We aim to design a framework for systematically designing verifiable credential schemas that satisfy complex requirements. To this end, we use partial graph modeling and graph generation [4] to generate diverse VC architectures, as the verifiable credential data model can be expressed as **subject-property-value** relationships, and thus verifiable credentials can be modeled as information graphs. To help ensure interoperability, the generated designs follow the model-driven architecture (MDA) approach [5].

After a brief introduction of verifiable credentials, we present our approach, its architecture, and its benefits. Finally, we discuss our future aims regarding our method.

2 Preliminaries

Verifiable credentials (VCs), as defined by the W3C [1], are a digital form of credentials with cryptographically verifiable authorship, making them more trustworthy and tamper-evident than their physical counterparts. They represent claims about a subject backed by their authors. These claims can be expressed as **subject-property-value** relationships, e.g. "Tom"-**has**-"blue eyes".

It is important to note that verifying a VC does not directly imply the truth of the claims encoded in it, only their authenticity and currency. Rather, after authenticity and currency

are established, a verifier may evaluate the claims based on their own policy and trust in the author.

VCs can be presented as *verifiable presentations (VPs)*, where multiple VCs are cryptographically linked together by the holder, meaning that not only the authenticity of the credentials is ensured, but so is the authenticity of the entity presenting them. VPs can also contain additional arbitrary data, for which one use-case links the aligned entities between related VCs.

In this model, the W3C identifies five roles:

- The **holder**, who possesses and presents VCs;
- The **issuer**, who asserts claims about one or more subjects, and creates VCs;
- The **subject**, about which claims are made;
- The **verifier**, who receives and processes VCs presented to them;
- And the **verifiable data register (VDR)**, which mediates the creation and verification of publicly used data, such as identifiers, *verifiable credential schemas*, revocation registries, and so on.

Verifiable credential schemas are data schemas that enforce a specific structure over data collection. *Data verification schemas* ensure that the structure and contents of a VC conform to a published schema. *Data encoding schemas* are used to map the contents of a VC to an alternative representation format.

VC schemas make establishing the structure of common claim sets easier; their design influences the privacy characteristics of a verifiable credential ecosystem. How the claims are divided up between credentials plays a crucial role in how expressive a set of related VCs is and how precisely a holder can control what they disclose when they present a VC or VP.

An important concept in this case is *Entity Alignment*, which is the process of identifying nodes that represent the same entity across multiple ontologies. In our case, if two VCs contain claims about the same subject, there needs to be a process that enables us to identify them and thus enable us to reason over the joint information graph, as shown in Figure 1.

Reasoning over information graphs constructed from verifiable credentials allows us to make inferences and evaluate policies over subjects in a privacy-preserving way. Although many small examples have been demonstrated in existing research, a systematic design approach has yet to be proposed for VC shema sets, over which a policy may reason.

3 Approach

To provide a systematic method for designing credential schemas which contain claims from a predefined ontology, i.e. which credential should contain which claim from an information graph model, we propose a *design space exploration (DSE)* [6] process using *partial graph modeling* [4], which produces templates for platform-specific verifiable credential schemas, over

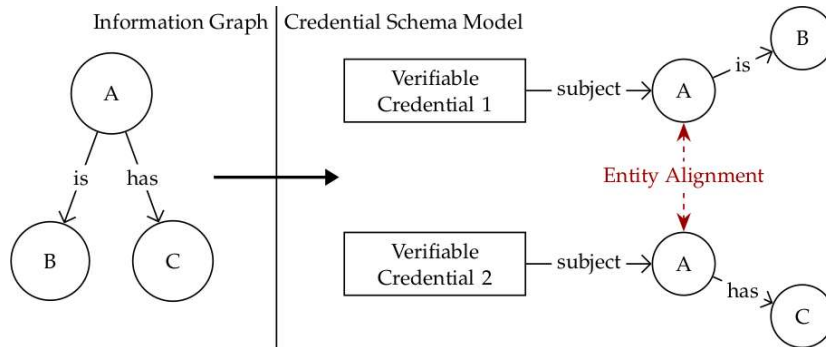


Figure 1: Decomposition of an information graph into verifiable credential schemas

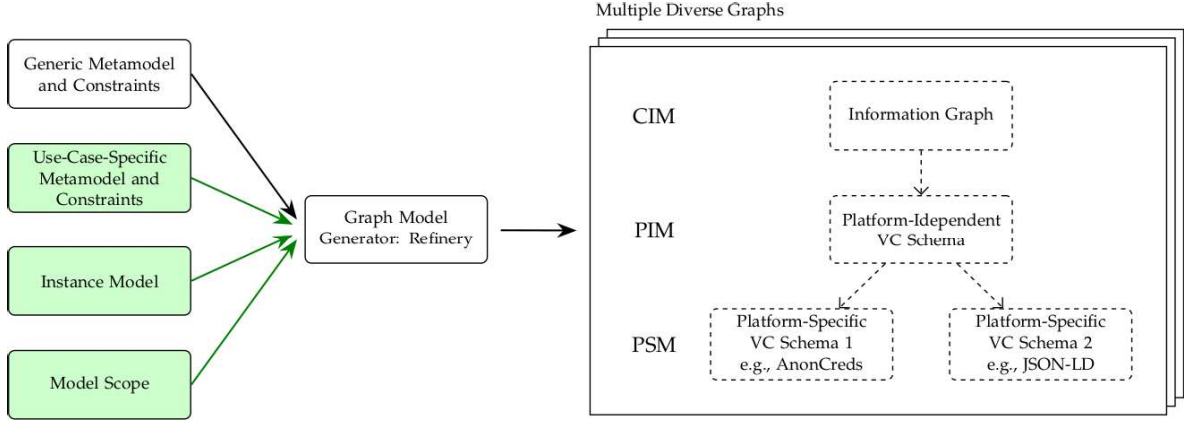


Figure 2: The architecture of model generation using Refinery [8]

which VCs can be issued, from which the required information graph can be constructed and reasoned over.

These designs follow the MDA [5] approach, as we recognize that interoperability is desirable between ontologies and diverse sets of VC schemas, as well as between platform-independent and platform-specific VC schema models, on platforms such as Hyperledger Anoncreds [7] and JSON-LD [2].

To generate our models, we use Refinery [8]¹, a partial graph refinement and graph generation tool, which can generate diverse graphs based on a ruleset written in its own domain specific language. It requires the following input:

- A **metamodel**, specifying nodes and edge types and how they can be connected;
- **Predicates**, which declaratively denote patterns in a given metamodel;
- **Error predicates**, which denote unwanted patterns, suitable for effectively shrinking the design space;
- **Instance models**, which are specific model parts that are required to be present in the generated model;
- and the **model scope**, which denotes the desired size range of the generated graph, with the option of including some specific model elements.

We define a generic metamodel and predicate set², which generates models that reflect the core logic of MDA. The generated model can be divided into three subgraphs analogous to the steps of MDA. Figure 2 provides an overview of the generated graphs.

The first is an **information graph**. This is the graph over which we want to later reason.

The relationships are then “partitioned” into claims within VCs in a traceable way, meaning there is a one-to-one correspondence between the relationships in the information graph and the claims they turned into within the credentials. This forms a **set of related VCs**.

Then, the abstract set of VCs is refined into **credential-format-specific representations**, from which they can be directly implemented. This ensures interoperability with multiple VC frameworks, making the final system design more versatile.

The model can be refined with use-case-specific metamodel elements, predicates, and an instance model. This means we can extend the model to satisfy complex requirements that can be stated as patterns on an information graph. Examples of such requirements include: the information graph should be a specific graph instance; two relationships must go into separate credentials; a credential must contain at most two claims; etc.

¹Online version of the tool is available at <https://refinery.services/>

²Current assets are available on GitHub at <https://github.com/BlackLight54/dse-vc-refinery>

Through this approach, a designer can generate diverse verification requirement setups for their specific use case, which satisfy complex requirements, and model an interoperable, potentially provably privacy-preserving credential ecosystem.

4 Conclusion

In this paper, we proposed a novel method for systematically designing verifiable credential ecosystems from information graphs. We presented our approach for generating diverse data presentation requirement sets in the verifiable credential context, utilizing partial graph modeling. Our approach follows MDA principles, facilitating not only systematic design but also technical interoperability.

Based on our initial proof-of-concept models, we are working on elaborating the full graph generation facilities required for handling representative, real-life scenarios. We also aim to create the methodology and tooling for a complete design workflow.

Acknowledgments

This work was partially created under, and financed through, the Cooperation Agreement between the Hungarian National Bank (MNB) and the Budapest University of Technology and Economics (BME) in the Digitisation, artificial intelligence and data age workgroup.

The work of Bertalan Zoltán Péter, partially supported by the Doctoral Excellence Fellowship Programme (DCEP) is funded by the National Research Development and Innovation Fund of the Ministry of Culture and Innovation and the Budapest University of Technology and Economics under a grant agreement with the National Research, Development and Innovation Office.

References

- [1] D. C. Manu Sporny, Dave Longley. Verifiable credentials data model v2.0. <https://www.w3.org/TR/vc-data-model-2.0/>.
- [2] G. Cohen. Verifiable credentials json schema specification. <https://www.w3.org/TR/vc-json-schema/>.
- [3] Essif lab. <https://essif-lab.eu/>.
- [4] M. Famelis, R. Salay, and M. Chechik. Partial models: Towards modeling and reasoning with uncertainty. In *2012 34th International Conference on Software Engineering (ICSE)*, pages 573–583, 2012.
- [5] S. J. Mellor, K. Scott, A. Uhl, and D. Weise. Model-driven architecture. In *International conference on object-oriented information systems*, pages 290–297. Springer, 2002.
- [6] E. Kang, E. Jackson, and W. Schulte. An approach for effective design space exploration. In *Proceedings of the 16th Monterey Conference on Foundations of Computer Software: Modeling, Development, and Verification of Adaptive Systems, FOCS'10*, pages 33–54, Berlin, Heidelberg, 2010. Springer-Verlag.
- [7] S. Curran, H. Yildiz, and S. Curren. Anoncreds specification. <https://hyperledger.github.io/anoncreds-spec/>, 2022.
- [8] K. Marussy, A. Ficsor, O. Semeráth, and D. Varró. Refinery: Graph solver as a service. *ICSE 2024 Demonstrations*, 2024.